

An exact finite-partition criterion for pointwise convergence into the regulated primitive space

Candidate full solution packet for arXiv:0911.2931

17 August 2026

Status. Full solution to the literal pointwise-convergence question, likely valid. The result is an intrinsic necessary-and-sufficient characterization; bounded literature searches found no later exact answer, but priority is not claimed.

1 Source question

Erik Talvila's *The regulated primitive integral* [1], arXiv:0911.2931, defines

$$\mathcal{B}_R = \left\{ F : \mathbb{R} \rightarrow \mathbb{R} : \begin{array}{l} F \text{ is regulated and left continuous,} \\ F(-\infty) = 0, \quad F(\infty) \in \mathbb{R} \end{array} \right\}.$$

Members are regarded as real-valued functions on $\overline{\mathbb{R}} = [-\infty, \infty]$, with the endpoint values equal to the respective limits. In the convergence section, after giving several sufficient conditions, the paper states on PDF page 36:

The Sorgenfrey topology of Section 10 makes each function in $\mathcal{B}_R$ continuous. However, no interval in $\overline{\mathbb{R}}$ is compact in this topology. Hence, Arzelà's theorem ([8 pp. 268]), establishing that quasi-uniform convergence is a necessary and sufficient condition for the limit of a sequence of continuous functions to be continuous, is not applicable. We do not know necessary and sufficient conditions under which a sequence in $\mathcal{B}_R$ will converge to a function in $\mathcal{B}_R$. However, Theorem 25 and Theorem 32 give a sufficient condition while Theorem 26 and Theorem 29 with their corollaries give conditions under which left continuity is preserved.

The central sentence asks for necessary and sufficient conditions for a sequence in $\mathcal{B}_R$ to converge to a function in $\mathcal{B}_R$. The surrounding section makes clear that the convergence at issue is pointwise on $\overline{\mathbb{R}}$. The theorem below gives a condition involving only the sequence, not an already known candidate limit.

2 Asymptotic mixed diameter

Let $\mathbf{F} = (F_n)_{n \geq 1} \subset \mathcal{B}_R$. For a nonempty $E \subset \overline{\mathbb{R}}$, define its *asymptotic mixed diameter*

$$\Delta_{\mathbf{F}}(E) := \sup_{x, y \in E} \lim_{N \rightarrow \infty} \sup_{n, m \geq N} |F_n(x) - F_m(y)|. \quad (1)$$

The inner limit exists in $[0, \infty]$, since the tail suprema decrease with N . The term mixed records that the two indices are independent. Most importantly, the index-tail limit is taken at fixed x, y *before* the supremum over the spatial set.

An *admissible partition* is a finite chain

$$-\infty = a_0 < a_1 < \cdots < a_q = \infty,$$

whose intermediate points are real. Its cells are

$$I_1 = [-\infty, a_1], \quad I_j = (a_{j-1}, a_j] \quad (2 \leq j \leq q). \quad (2)$$

The right-closed convention is the one naturally adapted to left continuity.

3 Proof intuition

At a fixed pair x, y , pointwise convergence $F_n \rightarrow F$ turns the two-index tail in (1) into $|F(x) - F(y)|$. Hence $\Delta_{\mathbf{F}}(E)$ becomes exactly $\text{diam } F(E)$. A left-continuous regulated function is precisely a uniform limit of finite step functions that are constant on cells of the form (2). This makes small mixed diameter on a finite half-open partition necessary.

For the reverse implication, a cell containing x bounds the tail diameter of the singleton $\{x\}$, so the condition forces $(F_n(x))$ to be Cauchy at every point. After taking the pointwise limit, the same cells have small ordinary diameter. Choosing value 0 on the first cell and the limit's value at each later right endpoint builds left-continuous regulated step functions converging uniformly to the limit. This simultaneously enforces the two endpoint limits, all right limits, and left continuity.

4 The exact criterion

We first isolate the standard half-open step characterization.

Lemma 1 (Half-open step criterion). *Let $F : \overline{\mathbb{R}} \rightarrow \mathbb{R}$ be real-valued and suppose $F(-\infty) = 0$. Then $F \in \mathcal{B}_R$ if and only if, for every $\varepsilon > 0$, there is an admissible partition whose cells satisfy*

$$\text{osc}(F; I_j) := \sup_{x, y \in I_j} |F(x) - F(y)| < \varepsilon \quad (1 \leq j \leq q).$$

Proof. Suppose first that $F \in \mathcal{B}_R$. A regulated function on the compactified line is a uniform limit of finite step functions [2, §7.13]; this is also the step-approximation fact used in [1]. Because F is left continuous, the values of a step approximant at its finitely many breakpoints may be replaced by the constants on their left. At the first cell one uses $0 = F(-\infty)$. Letting points approach a breakpoint from the left shows that this modification does not increase the limiting approximation error. After refining at all breakpoints, we obtain a left-continuous step function $S \in \mathcal{B}_R$, constant on cells of the form (2), with $\|F - S\|_\infty < \varepsilon/3$. Thus

$$\text{osc}(F; I_j) \leq 2\|F - S\|_\infty < \varepsilon.$$

Conversely, choose such a partition with cell oscillations below ε . Define a step function by

$$S_\varepsilon(x) = \begin{cases} 0, & x \in I_1, \\ F(a_j), & x \in I_j, \quad 2 \leq j \leq q. \end{cases}$$

Since $-\infty \in I_1$ and $a_j \in I_j$, the oscillation bounds give $\|F - S_\varepsilon\|_\infty < \varepsilon$. The half-open convention makes S_ε left continuous; it is regulated, equals 0 at the left endpoint, and is constant on a terminal neighborhood of $+\infty$. Thus $S_\varepsilon \in \mathcal{B}_R$. Taking $\varepsilon \downarrow 0$, uniform convergence preserves left continuity and transfers the Cauchy criteria for right limits and endpoint limits from S_ε to F . Therefore $F \in \mathcal{B}_R$. $\square$

Theorem 2 (Necessary and sufficient sequence criterion). *For a sequence $\mathbf{F} = (F_n) \subset \mathcal{B}_R$, the following are equivalent.*

- (i) F_n converges pointwise on $\overline{\mathbb{R}}$ to a function $F \in \mathcal{B}_R$.
- (ii) For every $\varepsilon > 0$, there is an admissible partition $(a_0, \dots, a_q)$ such that

$$\Delta_{\mathbf{F}}(I_j) < \varepsilon \quad (1 \leq j \leq q).$$

Proof. (i) implies (ii). Fix $x, y \in \overline{\mathbb{R}}$. Independent tail convergence at these two points gives

$$\lim_{N \rightarrow \infty} \sup_{n, m \geq N} |F_n(x) - F_m(y)| = |F(x) - F(y)|. \quad (3)$$

Indeed, the upper bound follows from

$$|F_n(x) - F_m(y)| \leq |F_n(x) - F(x)| + |F(x) - F(y)| + |F(y) - F_m(y)|,$$

and the reverse bound follows by applying the same inequality with the roles of the limit and the approximants reversed. Consequently,

$$\Delta_{\mathbf{F}}(E) = \text{diam } F(E) \quad \text{for every nonempty } E \subset \overline{\mathbb{R}}. \quad (4)$$

Lemma 1, applied with the prescribed ε , now gives the required partition.

(ii) implies (i). Fix $x \in \overline{\mathbb{R}}$. For every $\varepsilon > 0$, choose the partition in (ii) and let I_j be its cell containing x . Then

$$\lim_{N \rightarrow \infty} \sup_{n, m \geq N} |F_n(x) - F_m(x)| = \Delta_{\mathbf{F}}(\{x\}) \leq \Delta_{\mathbf{F}}(I_j) < \varepsilon.$$

Thus $(F_n(x))_n$ is Cauchy, and the completeness of $\mathbb{R}$ defines a pointwise limit

$$F(x) := \lim_{n \rightarrow \infty} F_n(x) \quad (x \in \overline{\mathbb{R}}).$$

In particular $F(-\infty) = 0$, because every $F_n(-\infty) = 0$. Identity (3), and hence (4), now holds for this F . Condition (ii) therefore gives, for every $\varepsilon > 0$, an admissible partition on whose cells $\text{osc}(F; I_j) < \varepsilon$. Lemma 1 yields $F \in \mathcal{B}_R$, completing the proof. $\square$

Remark 3 (Metric-valued strengthening). *Nothing in the proof uses linearity of the range. With absolute value replaced by a metric, the theorem remains true for functions taking values in any complete metric space, with $F(-\infty)$ fixed at a chosen basepoint. In particular it holds for Banach-space-valued regulated primitives.*

5 Why the order of operations is sharp

Define

$$F_n(x) = \mathbf{1}_{(0,1/n]}(x), \quad x \in \mathbb{R}, \quad F_n(\pm\infty) = 0.$$

Each F_n is regulated and left continuous, hence belongs to $\mathcal{B}_R$, and $F_n(x) \rightarrow 0$ at every $x \in \overline{\mathbb{R}}$. Therefore $\Delta_{\mathbf{F}}(E) = 0$ for every nonempty E , so Theorem 2(ii) holds even with one cell.

This example rules out several tempting stronger replacements:

- $F_n(0+) = 1$ for every n , so convergence of right traces is not necessary.
- No tail of the sequence is equiregulated at 0.
- If the spatial supremum is moved *inside* the index tail,

$$\widehat{\Delta}_{\mathbf{F}}(E) := \lim_{N \rightarrow \infty} \sup_{\substack{n, m \geq N \\ x, y \in E}} |F_n(x) - F_m(y)|,$$

then every finite admissible partition has a cell containing a positive right-neighborhood of 0, and $\widehat{\Delta}_{\mathbf{F}}$ equals 1 on that cell. Indeed, for large n , take $x = 1/(2n)$, $y = 2/n$, and $m = n$.

Thus the index-limit-first ordering in (1) is not cosmetic; it is what admits moving or shrinking boundary layers while still detecting the regularity of the pointwise limit.

The source paper's escaping-mass example is also detected exactly. If $F_n(x) = H_1(x - n)$, then the pointwise values are 0 at every finite x but $F_n(\infty) = 1$. On the terminal cell $(a_{q-1}, \infty]$, equation (1) compares a finite point with ∞ and gives mixed diameter at least 1; condition (ii) fails.

6 Integral formulation

Corollary 4. *Let $f_n \in \mathcal{A}_R$ have primitives $F_n \in \mathcal{B}_R$. There is an $f = F' \in \mathcal{A}_R$ such that*

$$\int_{(-\infty, x)} f_n \longrightarrow \int_{(-\infty, x)} f \quad \text{for every } x \in (-\infty, \infty]$$

if and only if the primitive sequence satisfies Theorem 2(ii).

Proof. Under the conventions of [1], $\int_{(-\infty, x)} f_n = F_n(x)$. The claim is therefore exactly Theorem 2, followed by $f = F'$. $\square$

7 Verification, scope, and novelty

Proof audit. The proof has only three structural inputs: completeness of the scalar range, the fixed-pair identity (3), and the elementary half-open step characterization in Lemma 1. The first and last cells include the endpoint values, so the criterion enforces the endpoint limits rather than merely local regulation. The right endpoint of each finite cell belongs to the cell on its left, which enforces left continuity while allowing an arbitrary right jump.

Scope. This is a full answer to the literal question of when a sequence in $\mathcal{B}_R$ converges pointwise on $\overline{\mathbb{R}}$ to a member of $\mathcal{B}_R$. It is an intrinsic characterization rather than a conveniently checkable domination, variation, or compactness hypothesis. Stronger sufficient conditions in the source remain useful precisely because they may be easier to verify.

Bounded novelty search. On 17 August 2026, the run registry, solution/attempt/proof-gap indexes, and the local arXiv corpus were searched by arXiv id and by combinations of the phrases pointwise convergence, regulated functions, equiregulated, quasi-uniform, bounded epsilon-variation, and necessary and sufficient. Web searches used the exact source title and the same close phrases. They found the source question, equiregulated sufficient criteria, and approximate-variation pointwise selection results, but no later paper stating an exact answer to this question. Because the result is elementary and intrinsic, an equivalent standard formulation may exist outside the search bounds; novelty and priority are therefore not asserted.

Human-review recommendation. Verify first the quantifier/order choice in (1), then the half-open endpoint conventions in Lemma 1. The shrinking-pulse example should be used as the quickest test that no hidden equiregulation or trace-convergence assumption has entered.

References

- [1] E. Talvila, *The regulated primitive integral*, Illinois J. Math. **53** (2009), no. 4, 1187–1219; arXiv:0911.2931.
- [2] R. M. McLeod, *The Generalized Riemann Integral*, Carus Mathematical Monographs 20, Mathematical Association of America, 1980.